

Original Research Article

Economic and environmental assessment of renewable energy micro-systems in a developing country

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ABSTRACT

This paper intends to evaluate the impact of the introduction of electricity generation micro-systems based on 100% renewable sources to supply electricity demands of non-electrified rural residential houses in Santiago Island's energy system (Cape Verde). For that purpose a comparison between three energy scenarios (2011–2030) were studied: a Reference Scenario, where it was assumed that the electricity grid would be expanded to increase households electrification rate and that grid connected renewable energy technologies will be used to increase generation power; a Diesel Scenario, where future electricity requirements would be provided by constructing new diesel power plants; and a Decentralized Scenario that is based on the Reference Scenario but that differs from it for the fact that it considers the installation of off-grid renewable energy micro systems (REMS) to attend electricity requirements of post-2010 electrified rural houses. The electricity supply–demand system of Santiago Island was modeled using an energy planning software and an optimal REMS design was modeled to provide electricity requirements of a typical Santiago's household. Although it is clear that using renewable energy technologies (Reference Scenario) presents better results from an environmental and economical point of view than installing new diesel power plants (Diesel Scenario), using off-grid REMS to electrify post-2010 rural households (Decentralized Scenario) result to be even more beneficial than generate and distribute on-grid renewable electricity.

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Introduction

Energy consumption is fundamental to sustain modern societies. It is needed to fulfil population's basic needs and to supply productive systems with the required power and heat. Over the years potential scarcity on energy resources has been solved by developing and using better technology to extract them. For a long period of time, per capita energy consumption was, in fact, a measure of nation prosperity: more energy used, more developed the nation would be. Energy planning could be summarized as the task of assuring supply of the correct amount of energy, on time and as inexpensive as possible [1,2].

The 1970's energy crisis changed that way of thinking: sources of fossil fuels were not as abundant, as available and as cheap as it was thought [3]. Simultaneously, environmental and social issues started to arise, namely the environmental degradation which is an outcome of burning fossil fuels and its effect on human health

[4]. Bottom line, economic criteria could not be sufficient to properly perform on energy planning. Energy planning must account for [4,5]:

- *Economic criteria:* Investment and operational costs.
- *Technical criteria:* Efficiency, Feasibility, Safety.
- *Environmental criteria:* Air pollution, resources scarcity, impacts on ecological habitats.
- *Social criteria:* Employment, Public acceptance, Public health.

Because testing and experimenting in real world situations is most of the times impractical, too expensive or impossible, energy models can be used to perform comprehensive calculations and system analysis in simplified representations of real systems. Jebaraj and Iniyar [6] studied a large variety of energy models, and grouped them in five categories: supply–demand models, forecasting models, optimization models, neural-network models, and emission models. Hiremath et al. [7] reviewed the use of energy models to perform decentralized energy planning and the integration of renewable sources of energy. The authors classified energy

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models as: optimization models, energy supply/demand driven models, resource energy planning models, and energy models based on neural networks. Urban et al. [8] approached the particularity of energy planning in developing countries, and studied the suitability of current energy tools to address the planning problems of those countries.

Using computer models to perform such experiments offers great advantages: it allows to explicitly state the assumptions that form the base of the model; the logical consequences of the modelers' assumptions are computed; many factors can be inter related simultaneously. Although some of the computer programs were developed for a specific energy system analysis, it is time-consuming to create new tools for each and every analysis. In that sense, several software tools were developed and used to create a model of any given energy system. Depending on the purpose of the analysis to be conducted, a suitable software tool must be selected. Connolly et al. [9] reviewed a large set of computer tools (37 to be precise) that could be used for energy planning allowing the integration of renewable energy into energy systems that would help the selection process of the "ideal" software for a specific situation and objectives.

Because of their economical, political and even geographical characteristics, developing countries' energy systems have specific features and obey to a particular rationality. This is even more perceived in islands due to their isolation, the lack of conventional energy sources and the small dimension of their energy market. Renewable energy systems (RES) have been presented as an interesting solution to provide electricity to isolated islands or remote villages [10]. A significant number of research papers that have addressed different RES configuration has been exploited. Some examples: stand-alone photovoltaic system [11]; a combined cooling, heating and power hybrid system [12]; a HES (Hybrid Energy System) comprised of photovoltaic panels and wind turbines grid-connected [13] or stand-alone [14]; and even wave energy systems have been studied [15]. While most of the research papers address the implementation of RES at a village and/or island scale, using individual house-level renewable energy micro systems (REMS) is also an alternative: a World Bank report [16] advocates the photovoltaic-diesel generator system as a good solution where customers are few and dispersed and their main electricity use is domestic lighting; in Peru, individual solar home systems were chosen as the T by the ACCIONA Microenergy Foundation (FUNDAME) to facilitate access to basic electricity services to rural communities with no expectation to get electricity networks [17]; for the rural Pacific island of 'O'ua, which is part of the Ha'apai Solar Electrification Project in Tonga, it was determined using least-cost analysis that solar home systems would provide the most cost-effective means of supplying electricity when compared with a village diesel generator for supplying basic household electricity services [18].

Cape Verde (a cluster of ten islands located on the North Africa's west coast) is one of the smallest and poorest "small island developing states" (SIDS) and has been used as case study in different analyses. Monteiro et al. [19] studied and analyzed the success of using renewable energy technologies in Santo Antão, a Cape Verde island. Their intention was to identify the best strategies for optimizing the energy supply mix in Cape Verde Islands. Duic et al. [20] analyzed different scenarios for the development of the Santiago Island's electricity system and studied how the Kyoto Protocol, through the Clean Development Mechanism, could influence the transfer of clean energy technologies. They concluded that although Santiago Island's electricity generation is heavily dependent on Diesel engines, expensive and polluting, it's still the most appropriate on such a small scale. The paper also shows that although the emission reduction on global scale is small, there is great potential for establishing a strong market presence of

renewable energy technologies in developing countries. The H₂RES model was used to design and analyze different scenarios with the objective of maximizing the penetration of renewable energies in the energy system of S. Vicente Island in Cape Verde [21]. The results show that it is possible to have more than 30% of yearly penetration of renewable energy sources in the electricity supply system, together with more than 50% of the water supplied to the population produced from wind electricity.

The purpose of this study is to assess, both economically and environmentally, the impact of a decentralized electrification plan, based on the expansion of electricity supply by installing renewable energy micro-systems (REMS) in a developing country (Santiago Island, Cape Verde). The following section will present the methodology used to perform the analysis. Santiago Island's electrical system and an optimal renewable energy micro system (REMS) to provide electricity requirements of a typical Santiago's household are designed and presented in 'Modeling Santiago Island Electricity System' and 'Design of a renewable energy Micro system (REMS) for a typical Santiago Island household' respectively. 'Santiago Island Energy Scenarios' will present the assumptions used to develop three different scenarios for the development of Santiago's electrical system. The results of both models and the analysis of the outcomes of all three scenarios will be debated on 'Results'. And, finally, conclusions will be pointed in 'Conclusion' of this paper.

Methodology

Connolly classified energy computer tools according to different criteria: type of calculations allowed by the tool (simulation, optimization, scenario analysis, etc.); the geographical area analyzed by the tool (global, national, regional, local, community); or the energy sectors included (transport, electricity, heat). To accomplish the purpose of this study the electricity system of Santiago Island must be simulated, an optimal REMS should be designed to attend the electricity requirements of a typical Santiago house, and the impact on the Santiago electricity system of introducing the optimal REMS design in several houses on the island must be assessed. By analyzing Connolly's review and having in mind the specific purpose of this paper, it was clear that a single software was not able to perform the analysis required: a simulation and scenario analysis at national level and an optimization analysis to a local level. Two software tools were used on this paper and a visual description of how the two software tools were integrated and used is presented in Fig. 1.

Long-range Energy Alternative Planning (LEAP) [22] uses an annual time-step, and the time horizon can extend for an unlimited number of years (typically between 20 and 50). It supports a number of different modeling methodologies: on the demand-side these range from bottom-up, end-use accounting techniques to top-down macroeconomic modeling; on the supply side, it provides a range of accounting and simulation methodologies for modeling electricity generation and capacity expansion planning. LEAP can simulate all sectors, all technologies, and all costs within an energy-system, as well as externalities for any pollutants, decommissioning costs, and unmet demand costs. LEAP also includes a scenario manager that can be used to describe individual policy measures. The resulting scenarios are self-consistent storylines of how an energy-system might evolve over time. LEAP displays its results as charts, tables, and maps which are user-defined and can be exported to Excel or PowerPoint. LEAP was used in this paper with two main purposes: (1) to design the electricity demand-supply system model of Santiago Island; (2) to develop three different scenarios for the energy model evolution, for a 20-year time horizon.

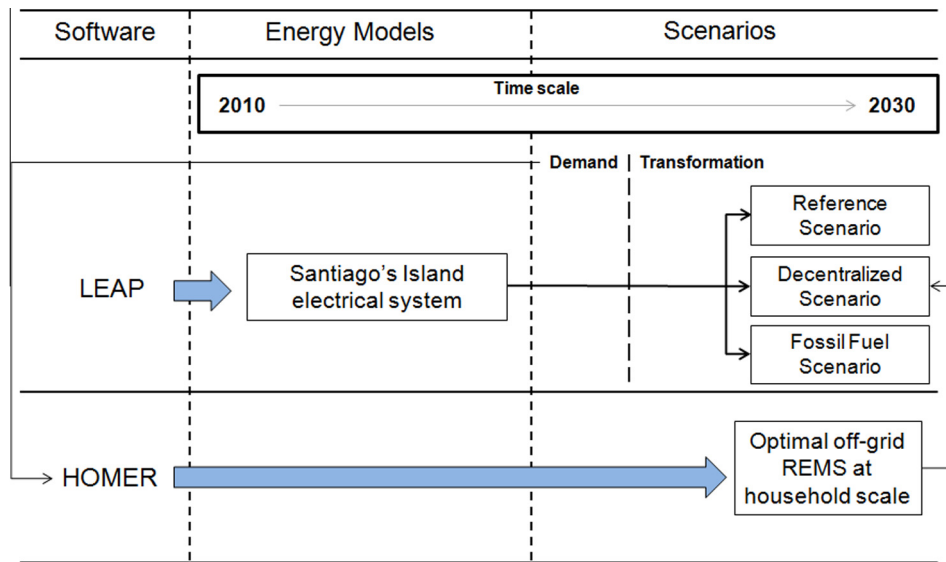


Fig. 1. Conceptual view of the methodology used.

HOMER [23] simulates and optimizes stand-alone and grid-connected power systems with any combination of wind turbines, PV arrays, run-of-river hydro power, biomass power, internal combustion engine generators, micro turbines, fuel cells, batteries, and hydrogen storage, serving both electric and thermal loads (by individual or district-heating systems). Also, all costs (including any pollution penalties) except fuel handling costs and taxes are included. The simulation considers a one-year time-period using a minimum time-step of 1 min. It performs a sensitivity analysis which can help the analyst to do ‘what-if’ analyses and to investigate the effects of uncertainty or changes in input variables. The objective of the optimization simulation is to evaluate the economic and technical feasibility for a large number of technology options, while considering variations in technology costs and energy resource availability. HOMER was used in this paper to find the optimal REMS configuration to provide a typical electricity annual requirement of a residential house in Santiago Island.

Modeling Santiago Island electricity system

A model of Santiago Island's electricity system was designed using LEAP. In order to do so several pieces of data regarding Santiago Island were collected, such as historical data on electricity consumption, GDP, population, number of electrified households, and typical household electrical consumption profile. The year 2010 was used as the base year to characterize its demand and supply sectors.

Cape Verde's population of less than a half million is spread over 9 islands. Santiago Island is the largest and the most populated island of Cape Verde, where the capital Praia is also situated (in 2010 about 56% of Cape Verde's total population lived on the island of Santiago [24]). Since its independence in 1975, the population has had an annual growth rate of 2.4% [25].

From the economic point of view it is a developing country experiencing constant growth. Between 2001 and 2007 GDP growth averaged 6.5% [25]. It does not have its own fuel resources and there is also a shortage of drinking water.

Electricity Demand

LEAP uses a calculation methodology that accounts for energy consumption as the product of an activity level (a measure of the

social or economic activity for which energy is consumed, such as household, person, GDP) and annual energy intensity (energy use per unit of activity):

$$\text{Energy Consumption} = \text{Activity Level} \times \text{Energy Intensity} \quad (1)$$

LEAP's demand analysis is a disaggregated end-use based approach for modeling the requirements for final energy in a particular area. Hence, electricity demand on Santiago Island was disaggregated on four different sectors: Residential, Governmental Entities, Water Desalination and Pumping, and Others (agriculture, business and industry). As far as demand modeling is concerned two different methodologies were employed: a “bottom-up” approach for the “Residential Sector”, and a “top-down” approach for the remaining sectors.

For each demand sector except “Residential” were used as input values the respective activity level and energy intensity while total energy consumption results from the multiplication of the previous two values – Table 1.

For the “Residential Sector” it was used “household” as the activity level. Using a “bottom-up” approach, a set of electricity consuming devices per household was considered. Table 2 presents information about the set of electrical devices on a Santiago Island household, their penetration rate, power, and hours of use [24]. Fig. 2 shows how the calculations were made to account for the total amount of electricity consumed at the “Residential Sector” using a bottom-up approach. It is clear that the total electricity consumption depends on devices' energy intensity, their saturation/penetration rate and the number of electrified households on both urban and rural areas (the rural and the urban population use electricity differently). Average household electricity consumption in 2010 is 3.8 kWh/day.

Results of the energy model shows energy usage in Santiago Island broken down by demand sectors (Fig. 3). Residential sector is responsible for the biggest share of electricity consumption. This fact is typical of a “developing country” like Cape Verde, while energy use in developed countries is very often dominated by the tertiary (services) and secondary (manufacturing) sectors of the economy. Because drinkable water is a scarce resource in Cape Verde, the energy used to produce clean water represents a significant share of total electricity used (more than 7%).

Table 1
Activity level and energy intensity of electricity demanding sectors.

Sector	Activity level			Total energy consumption		Energy intensity (kWh/Activity Level/year)
	Unit	Quantity	Source	Quantity (kWh)	Source	
Governmental Entities	Population	273 912	[25]	51 221 544	[26]	187
Water desalination	m ³ of distributed water	2509015	[26]	10286962		4.1
Others	GDP (US \$) <i>per capita</i>	1904	[27]	40079200		21 050

Table 2
Set of electrical devices used on Santiago's houses and their penetration rate, power, and time of use.

Device	Penetration rate (%)		Power (W)		Using hours (h/year)	
	Urban	Rural	Urban	Rural	Urban	Rural
60 W Light bulbs	4 ^a	2 ^a	60	60	1460	1460
40 W Light bulbs	4 ^a	4 ^a	40	40	1460	1460
Fridge	69.47	35.26	130		3650	
TV	82.79	56.64	80	60	1825	
Micro-waves	27.63	8.92	1200		62	44
PC	28.74	8.56	120		1825	1825
Video/DVD	63.00	39.94	35		146	120
Iron	89.00	58.24	1000		73	44
Wash machine	22.70	4.05	500		219	219
Air conditioned	4.08	1.32	1000		2190	2190
Freezer	12.83	7.02	200		1825	1825
Stereo	30.76	13.26	40		1095	1095

^a Number of light bulbs.

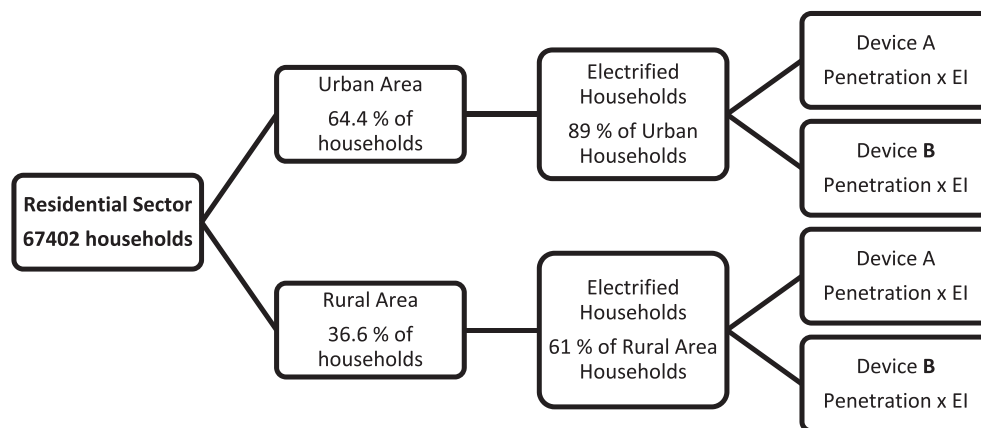


Fig. 2. Visual explanation of the methodology used to assess electricity use at Santiago's Island residential sector.

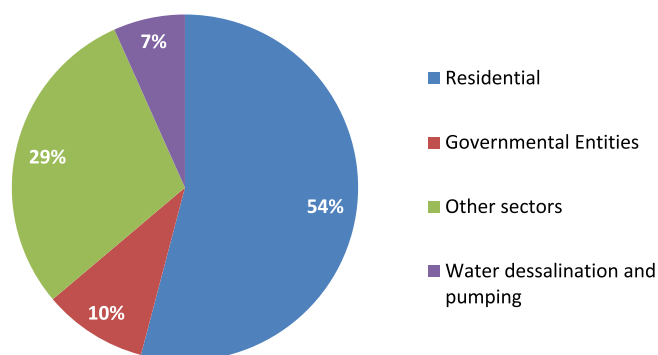


Fig. 3. Electricity use broken down by demand sector.

Electricity Supply

On 2010, Santiago Island had nearly 47 MW of electricity generation power and it was shared by six diesel power stations, one

wind farm, and one photovoltaic farm resulting in an annual production of 177 642 MWh (Fig. 4). Each of the power plants has different capital and variable/fixed operation and maintenance (O&M) costs.

Table 3 presents the operational characteristics assumed for each power plant type and their running costs. Electricity losses on the distribution grid were as high as 20% in 2010 [26].

Design of a renewable energy Micro system (REMS) for a typical Santiago Island household

The “Decentralized Scenario” includes the installation of individual REMS in all houses electrified after 2010. Because of that, optimal REMS was designed to provide electricity needs of a typical Santiago Island household. Furthermore, all three scenarios imply electricity load increase of the typical household over the simulation period (2010–2030). Consequently, the optimal REMS system was simulated in order to provide not the 2010 electricity load (4.14 kWh/day and 2.74 kWh/day, on urban and on rural area

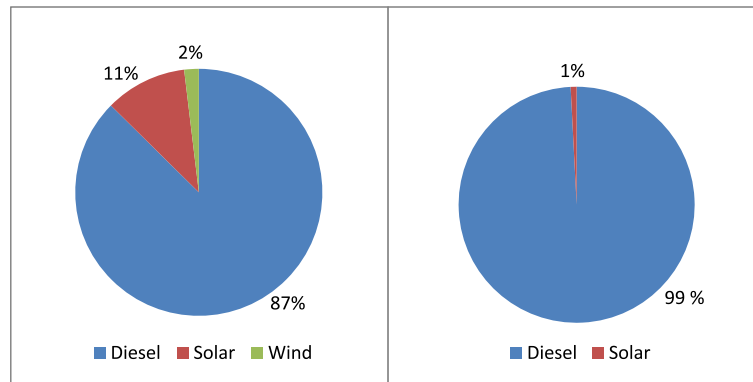


Fig. 4. (a) Total electrical power generation on Santiago Island, in 2010, broken down by technology, and (b) share of total electricity generated by each technology in 2010 [26].

Table 3

Technical and economic characteristics of the processes used for electricity generation on Santiago Island.

	Diesel	Wind	PV
Power capacity (MW)	41.83	0.97	5.17
Process efficiency (%)	40	100	100
Power plant availability (%)	80	30	30
Capital cost [€/kW]	600 ^b	1200 ^b	3800 ^b
O&M fixed cost [€/MW]	150.4 ^b	–	42.4 ^b
O&M variable cost [€/kWh]	–	0.019 ^b	–

^b Ref. [26].

respectively) but the average 2030 household load (7 kWh/day, value that results from the assumptions made for the demand forecast on ‘Electricity demand’). The lack of data on the typical electricity load profile of a typical house in Cape Verde was mitigated by using the load profile of a typical African house [28], presented on Fig. 5.

REMS should be based in solar and/or wind energy, so its optimal configuration depends on solar and wind availability. As a result, wind speed and solar radiation average values were considered to perform the simulation (Fig. 6). Norgard and Fonseca [29] assessed average wind speeds on Santiago Island and data on solar radiation is available on NASA’s database [30] (which can be downloaded directly from the software). HOMER software computes wind speeds for each of the 8760 h of the year using average wind speed on a monthly basis and two other parameters relating to seasonal and daily patterns – Weibull distribution and autocorrelation factor.

HOMER compares multiple technology options for a power system design. In this case, HOMER will simulate systems that include

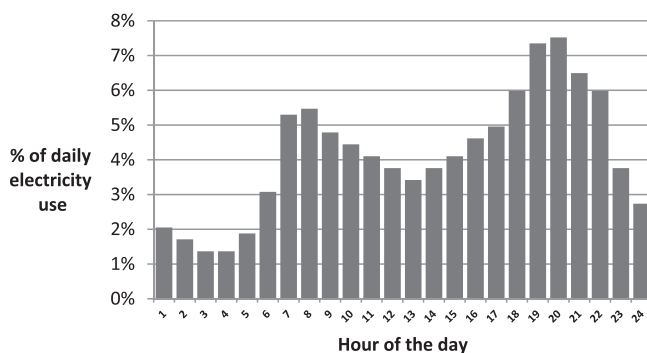


Fig. 5. Hourly distribution of electricity use at a typical Santiago Island house.

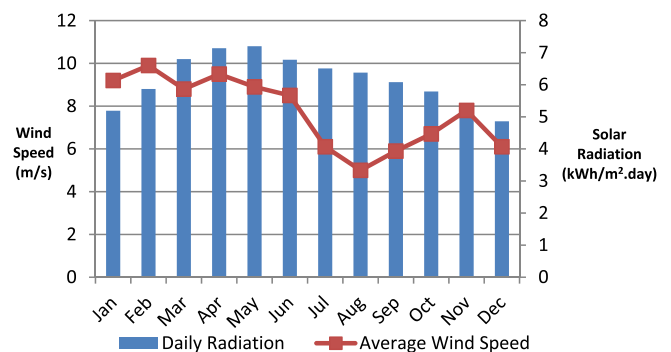


Fig. 6. Average solar radiation and wind speed on Santiago Island.

a wind turbine and/or photovoltaic panels, a battery-set and a converter/inverter (Table 4). The components used in simulation were selected from the internal database of the HOMER software. By providing several options for the number and/or power of system’s components, HOMER will find the technical and economically optimal REMS design to provide a typical 2030 household electric load, subject to the availability of the renewable resources. Because electricity storage capacity should bear at least 3 days of average load (approximately 21 kWh), this parameter will work as a constraint.

Santiago Island energy scenarios

A set of three scenarios was developed for a 20-year time frame (2010–2030). Electricity requirements will evolve identically in all three scenarios during the time frame while the electricity generation processes will be distinct:

- (1) *Reference Scenario*: The Cape Verde actual energy policy for the development of its electricity system. It is based on the expansion of the electric generation and distribution system through the installation of new renewable energy technologies (wind and PV) power stations connected to the central grid.
- (2) *Diesel Scenario*: The electricity generation system would be expanded by installing new diesel power plants.
- (3) *Decentralized Scenario*: It differs from the Reference Scenario because in this scenario the increase on electricity requirements from the residential sector will be provided by small decentralized electricity production systems.

The following section will address the assumptions adopted in order to simulate the evolution of the demand side (common to all three scenarios) and of the supply side of all three scenarios.

Table 4

Components used to simulate an optimal RES system for a typical house on Santiago Island.

	Denomination	Power/Number	Capital cost [*] (€)	O&M cost [*] (€/Year)
Wind turbine	SW Whisper 100	0.9 kW	2400	24
PV panel	BP 4180 T	0.180 kW	446	4
Battery	Surrete 4KS25P	3	3048	0
Inverter	AJ 275 S	0.275 kW	272	0

^{*} All costs from Ref. [31].

Demand

Residential sector

In this particular sector electricity demand will be determined over time by: total number of households at the island; social mobility, which results from people moving from rural to urban areas; houses electrification rate, both on rural and urban areas; penetration/diffusion rate of electricity-consuming devices on both rural and urban areas.

Santiago Island's demographic evolution was based on the United Nations moderate forecast (increase of 1.03%/year) [32]. It was also assumed that a typical household consists of 4 people and that it remains unchanged during the simulation period. Social mobility forecast (rural to urban areas) was performed using a linear forecast based on historical data. Electrification rate followed a simple logistic forecast model [33]; the three-parameter S-shaped logistic growth model (Eq. (2)). This forecast model is applied to socio-technical systems in order to describe continuous “trajectories” of growth or decline through time.

$$N(t) = \frac{\alpha}{1 + e^{(-kt - \beta)}} \quad (2)$$

Eq. (2) states that penetration rate is function of: k = scale factor, β = specifies the time (t) when the curve reaches 0.5 K; α = maximum penetration rate. Results from the forecast are shown in Table 5.

It is assumed that average electricity consumption per household will rise around 75% from 2010 until 2030: 7 kWh/day. This is the electricity requirement used to design the optimal REMS (‘Methodology’).

Governmental sector

As already stated on Table 1, electricity consumption of governmental entities is directly related to the population of the island. This assumption is validated by the linear regression made to historical data (2005–2010) on population [25] and government entities electricity consumption [26], which resulted in a high coefficient of correlation (R^2 of 0.95). It was also assumed that the energy intensity (kWh/person/year) at this sector will remain steady for the entire simulation period. As stated in Section 5.1.1, Santiago Island's population forecast assumed in this paper is based in United Nations data [32].

Business/Industry/Agriculture

For all three business, agriculture and industry sectors it was assumed that electricity consumption is directly dependent of GDP *per capita*. Once again, this assumption is validated by the linear regression made to historical data (2005–2010) on GDP *per*

capita [27] and electricity consumption of primary, secondary and tertiary economic activities [34] (R^2 of 0.94). Energy intensity will remain steady over the simulation period. Concerning the expected evolution of the GDP per capita, it was performed a linear forecast based on a regression analysis using data from the 1990 to 2009 period [27] (R^2 = 0.82).

Water desalination and pumping

The scarcity of natural fresh water sources on Cape Verde, a key feature to the country's economic development, has been sheepishly overcome through sea water desalination. As a result, water desalination plays a major role on Cape Verde's development and is an expensive and energy intensive process. On 2010, 88% of all distributed water on Santiago Island was provided by desalination water stations [26]. This framework results on a low quality water supply system enabling no more than 25 liters of water consumption per capita per day. According to the World Health Organization standards, an optimal water supply system should provide at least 100 liters per capita per day [35]. For simulation purposes it was assumed that water supply will increase linearly until it reaches 100 liters per capita per day on 2030, and the energy intensity (kWh per cubic meter of water desalinated) will remain steady throughout that period.

Supply

As described at the beginning of this section, the three energy scenarios developed for the island distinguish themselves by the processes used to generate electricity. Thus, the assumptions adopted for each scenario are described in the following paragraphs.

Reference Scenario

This energy scenario adopts the following assumptions:

- (1) It reflects Cape Verde's governmental policy, which aims to achieve 50% of renewable electricity (PV and wind) in 2020 and to maintain this share until 2030.
- (2) All the electricity (renewable and non-renewable) will be generated on power plants, and distributed through a centralized distribution grid.
- (3) Electricity grid losses will be of 20% through the 20-year period.
- (4) The system reserve margin shouldn't be lower than 65%:

$$\text{Reserve Margin} = \frac{\text{power capacity} - \text{peak load}}{\text{peak load}} \times 100 \quad (3)$$

- (5) House electrification and connection to the electricity grid costs 2256€/house.
- (6) All electricity requirements must be provided.

Diesel Scenario

This scenario is similar to the previous one, except from two facts: no additional renewable power plants will be implemented until 2030, and the reserve margin must remain above 30%. This

Table 5

Forecast for the evolution of the household stock on Santiago Island.

	2010	2030	Type of Forecast
Total households	67 032	82 160	Linear
% Urban/% rural	64.4%/33.6%	75.1%/24.9%	Linear
% Electrified urban houses	89%	99.5%	Logistic
% Electrified rural houses	61.3%	97.2%	Logistic

means that new diesel power plants added will run on diesel in order to maintain the reserve margin above 30%.

Decentralized Scenario

It inherits the assumptions made for the Reference Scenario. In addition, it is assumed that all rural houses electrified from 2010 onwards will not be connected to the grid. Instead, they will have REMS to generate electricity, according to the assumptions stated on 'Design of a renewable energy Micro system (REMS) for a typical Santiago Island household'.

Fuel costs

Cape Verde's electricity system is highly dependent on imported oil derivatives. Thus, evolution of oil prices during the simulation years (2010–2030) will have a great impact on the simulation results. According to the US Energy Information Administration [36], a reference scenario for oil price evolution will presuppose the trend shown on Fig. 7. It was assumed that the price of imported oil derivatives to Cape Verde will adopt the same trend.

Results

Optimal REMS design for a typical household

The optimal REMS configuration suitable for the electricity demand of a typical Santiago Island house in 2030 resulted to be a 2.52 kW PV system, with 3 batteries (21 kWh of usable nominal capacity, meaning 72 h of autonomy), and a 1.1 kW converter. This REMS has an initial capital cost of 12400€ and O&M cost of 177€/year. REMS' energy behavior can be analyzed on Table 6 and Fig. 8. Electricity requirements of a typical house are primarily provided by the PV array. Electricity production surplus is stored in batteries. However, the storage capacity is limited and electricity may be wasted. The storage capacity never drops below 40% of total capacity (Fig. 9).

Scenario analysis

Demand

The assumptions adopted for the present analysis and stated on 'Santiago Island Energy Scenarios', will result on a threefold increase of electricity demand by 2030. Despite all sectors would require more electricity on 2030, fresh water production and distribution would have a major weight on total electricity consumption in 2030: to achieve a decent fresh water distribution system will require almost 30% of all electricity consumption on 2030 (Fig. 10).

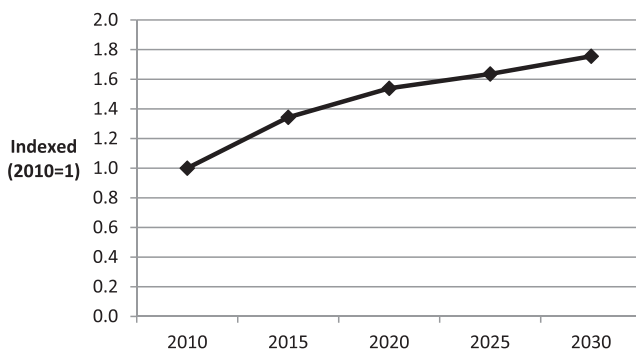


Fig. 7. Projection for world oil price on a reference scenario according to the US Energy Information Administration [36].

Table 6

Production, use and waste of electricity for the optimal REMS configuration.

Component	kWh/year
PV array total production	4517
Typical house load	2554
Electric requirements provided by batteries	1507
Electric requirements provided by PV array	1047
Wasted electricity	1317

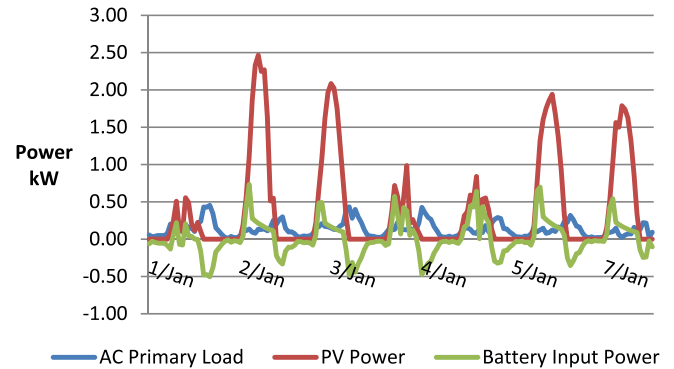


Fig. 8. Energy behavior for the optimal REMS configuration during a week-period.

Regarding the residential sector, and despite the assumptions undertaken (social mobility, population growth and increase on electric devices), electricity consumption share on urban and rural areas will remain steady during the simulation time period (approximately 20% rural for 80% urban). Overall use of electricity in a typical household would remain similar from 2010 to 2030, while AC use can see a significant increase (Fig. 11).

Supply

As already stated on 'Santiago Island Energy Scenarios', the processes used to supply electricity will differentiate the three energy scenarios: Reference scenario, Decentralized scenario, and Diesel scenario. This last one is used mainly to allow a proper assessment of the actual Cape Verde's energy policy (Reference Scenario).

A logical outcome from assuming that part of the future residential electricity requirements would be provided by REMS isolated from the central grid would be a decrease of total wasted electricity distributed by the grid (Fig. 12). Assuming that grid losses remain constant throughout the simulation period, the Reference Scenario would waste 20% of all electricity generated at power stations. Decentralized Scenario, on the other hand, presumes a growing amount of electricity that will provide the residential demand without being distributed by the central grid, resulting on a lower amount of total electricity wasted (on 2020 only 10% of the total electricity generated on Cape Verde's system is wasted).

In order to provide the electricity demand and simultaneously maintain the system's reserve margin (65%) both Reference and Decentralized Scenario will have power capacity increase. Because the amount of electricity distributed by the central grid is lower on the decentralized scenario, the total amount of added capacity is correspondingly lower (Fig. 13). The total amount of electricity delivered by the set of installed REMS grows every year at the Decentralized Scenario (Fig. 14).

Overall scenario assessment can be performed analyzing Table 7. Options undertaken by the current Cape Verde's energy policy will result on significant better economical and environmental results when comparing to a scenario with a fossil fuel-

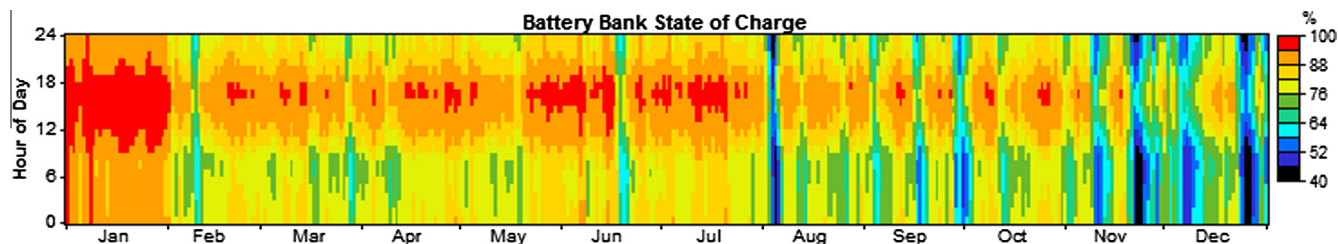


Fig. 9. Yearly variation of the battery bank state of charge.

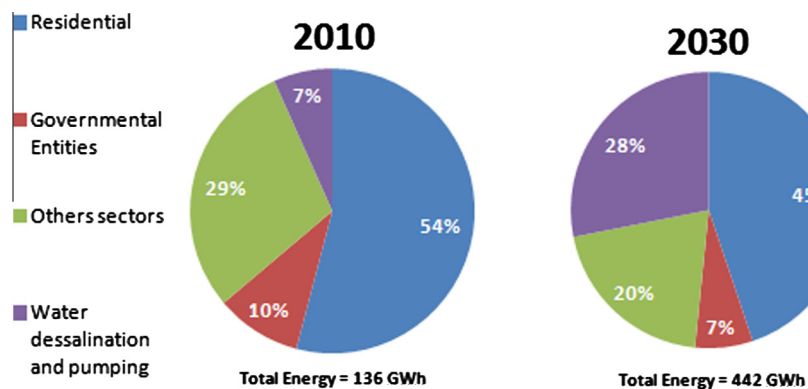


Fig. 10. Electricity use in 2010 and 2030.

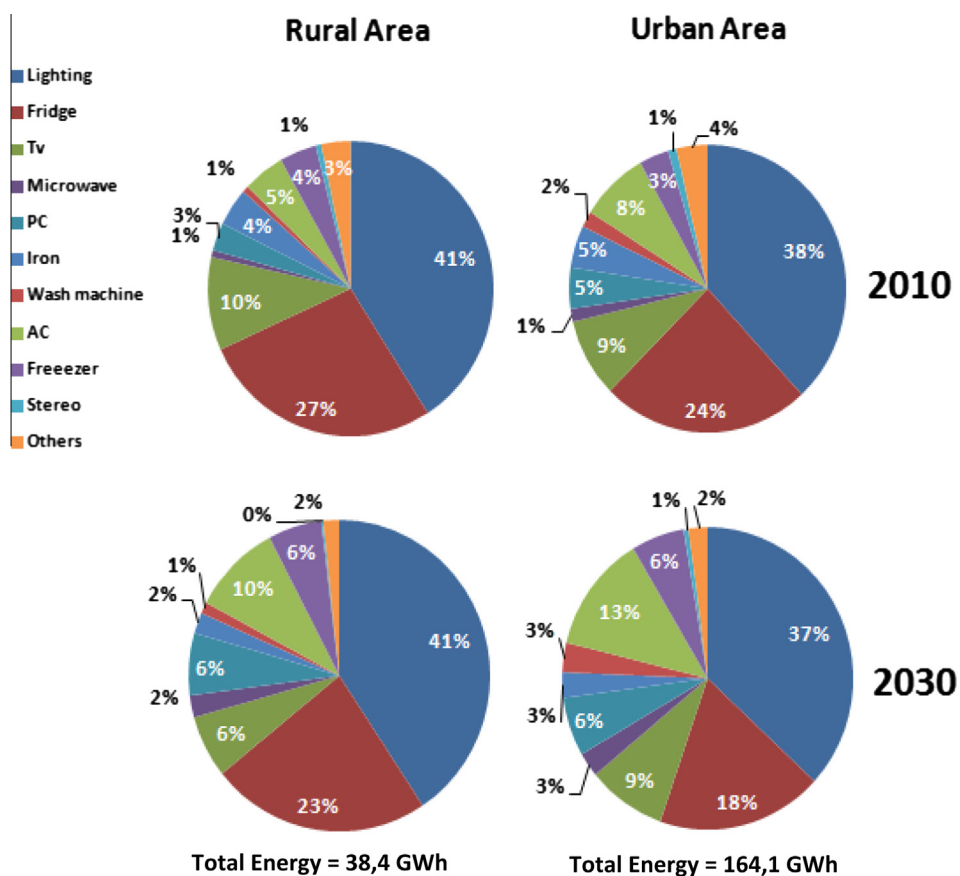


Fig. 11. Electricity use broken down by electrical devices in 2010 and 2030, both on rural and urban areas.

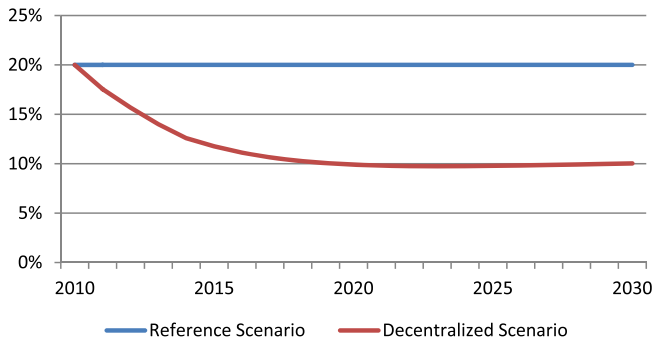


Fig. 12. Share of wasted electricity out of the total amount of grid-distributed electricity.

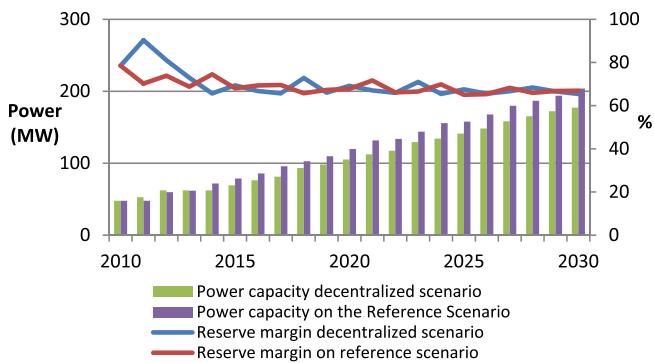


Fig. 13. System's reserve margin and total power capacity for the grid-connected electricity system.

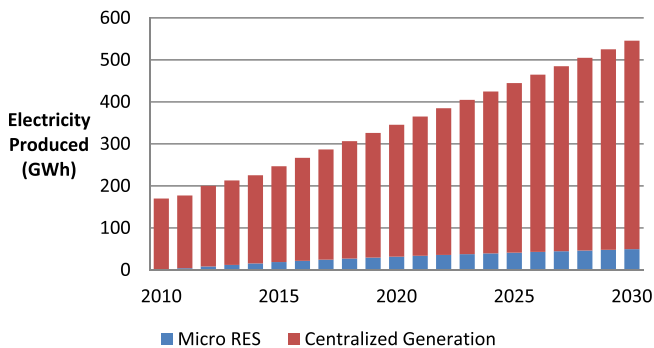


Fig. 14. Total electricity generated on the Decentralized Scenario disaggregated by type of electricity: electricity from the grid and electricity generated at the house's REMS.

Table 7

Economical, environmental and resource import performance of each scenario.

Scenario	Net present cost (M€)	CO _{2eq} emissions (kTon)	Diesel imports (KTOE)
Reference	1737	628	207
Fossil fuel	2060	916	301
Decentralized	1632	576	190

based electricity system. However, implementing decentralized off-grid REMS presents even better results.

On Fig. 15 one can acknowledge economic results of both Diesel and Decentralized Scenarios against the Reference Scenario. Diesel Scenario will not need as much power capacity as the Reference

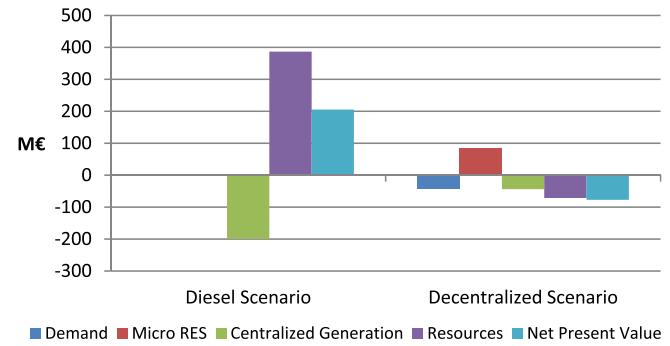


Fig. 15. Cost-benefit analysis of both Diesel and Decentralized scenarios regarding the Reference Scenario.

Scenario, so the costs on this particular area will be significantly lower. However, over a 20 year horizon, the money spent on importing diesel on a scenario fully relying on fossil fuel technology would overcome the savings made by not investing on new renewable power plants. Installing off-grid REMS would raise the costs of total electricity generation capacity but on the other hand it lowers expenditures on importing fossil fuels. Thus, savings from not connecting new houses to the central grid results on a lower Net Present Cost in Decentralized Scenario.

Conclusion

Identifying renewable energy sources and assessing their potential have great interest in today's energy planning process. Given their diffuse nature, intermittency, specific technological maturity and market situation, renewable energy planning process is totally different from traditional energy planning. A greater integration of renewable energy technologies requires a shift from the traditional procedure of planning and organizing energy systems. This can be achieved by a better choice of decision criteria which will influence decision process on energy planning.

As part of a developing country, it is expected that the overall electricity demand in the Santiago Island will rise significantly over the next years. Two sectors will have an important role on increase of electricity demand: residential and water desalination sectors. The increase on residential sector is mainly related to the significant share of residential houses that were not electrified in 2010, mainly in the rural areas, and that would be electrified in 2030 (almost 100% of all houses will be electrified). The provision of fresh water is fundamental to improve the basic standard of living. Fresh water is a scarce resource on Santiago Island in 2010, below the minimum acceptable value according to UN guidelines, and the water desalination processes are the only viable option to provide the required fresh water volume. This is however an energy intensive process and therefore energy consumption in this particular sector is expected to rise more than 10 times between 2010 and 2030.

Results from this paper show that increasing power capacity by installing new fossil fuel based power stations results to be the worst option from both environmental and economical points of view: Diesel Scenario presents the highest value of CO_{2e} emissions and it is the most expensive scenario (highest Net Present Value). Reference and Decentralized scenarios present close results, although electrifying rural houses by installing stand-alone REMS turned out to be a better option than investing on a centralized renewable electricity production and distribution. Despite electricity generated through REMS would only account for 9% of total electricity production in 2030, it would be enough to downsize electricity waste on the distribution grid to near 10% of total electricity generated.

It was assumed that in order to guarantee minimum electricity supply safety, stand-alone REMS should include a three day storage capacity. Under the simulated situation, stored electricity per house wouldn't reach less than 40% of maximum storage capacity at any moment of the year, and over more than 90% of the time would be over 75% of total storage capacity.

Results from the scenarios comparison suggest the relevance of assuming the possibility of implementing off-grid individual REMS to generate electricity instead of expanding the electricity grid of a centralized system to generate and distribute electricity in developing countries with a scarce and insufficient electricity distribution grid, dispersed population and abundant renewable resources. When assuming the increase of electricity demand presented in this study it is also evident that the bold target from the Cape Verde government to achieving 20% renewable electricity in 2020 is economically appropriate in a 20 year time-frame.

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